Faculty of Computing

Year 2 Semester 1 (2026)

IT2011 - Artificial Intelligence and Machine Learning

Group Assignment Specification

Group Size: 6 Members

Assignment Title:

Design, Implement, and Evaluate AI/ML Solutions for a Real-World Problem, Using a given Dataset

Objective
To design and implement an end-to-end AI/ML solution for a real-world problem using suitable supervised or unsupervised learning algorithms, applied to a dataset assigned to your group, and to critically evaluate the performance and ethical implications of the chosen approach.

Learning Outcomes Addressed:

- LO3: Apply AI/ML techniques to solve real-world problems
- LO4: Critically evaluate the effectiveness and impact of AI/ML algorithms

Assignment Components:

Step 1: Assigned Dataset Understanding and Problem Framing (10 Marks)

- Each group will be provided with a specific dataset by the module coordinators. Dataset selection is not required this year , your task is to understand, describe, and justify the suitability of your assigned dataset.
- Identify a real-world problem domain (e.g., healthcare, finance, agriculture, education) that aligns with the assigned dataset.
- Describe and justify the dataset's suitability (features, labels/target variable, size, and quality).
- Briefly state which AI tools (if any) you plan to use during the project and for what purpose (see the AI Tool Usage note below).

Step 2: Data Preprocessing and Analysis (25 Marks)

- Clean and preprocess the assigned dataset (e.g., missing values, outliers, encoding).
- Perform exploratory data analysis with visualizations.
- Provide insights and patterns observed in the data.

Step 3: Model Design and Implementation (25 Marks)

- Select and justify at least two AI/ML algorithms (e.g., SVM, Decision Trees, K-Means, Neural Networks) suitable for the assigned dataset.
- Implement models using Python (Pytorch, TensorFlow, etc.).
- Include hyperparameter tuning and model pipeline steps.

Step 4: Model Evaluation and Comparison (30 Marks)

- Use relevant performance metrics (Accuracy, F1 Score, Confusion Matrix, etc.).
- Perform k-fold cross-validation.
- Compare models and justify the best performer.

Step 5: Group Report (10 Marks)

a) Prepare a professional report (max 15 pages):

1. Introduction and Problem Statement
2. Dataset Description (assigned dataset: source, structure, and characteristics)
3. Preprocessing & EDA
4. Model Design and Implementation
5. Evaluation and Comparison
6. Ethical Considerations and Bias Mitigation
7. AI Tool Usage Declaration and Transparency (new)
8. Reflections and Lessons Learned
9. References

b) Submit your report in PDF format before the viva session.

c) Use of visuals, teamwork, and clarity will be evaluated.

AI Tool Usage Declaration - what to include

We would like each group to be transparent about how generative AI tools (e.g., ChatGPT, Claude, Copilot, Gemini, etc.) were used during the project. In the dedicated report section, include a short table or write-up covering, for each tool used:

- The tool name and version (if known).
- The specific aspect(s) it supported - e.g., code debugging, EDA/visualization ideas, hyperparameter search, writing/editing the report, generating explanations, brainstorming problem framing, etc.
- The extent of use for each aspect (e.g., minimal / moderate / extensive).
- How the group verified, tested, or took ownership of AI-assisted outputs before including them.

Groups that used no AI tools should simply state this clearly. Honest disclosure is what is being assessed here, not the amount of AI use itself.